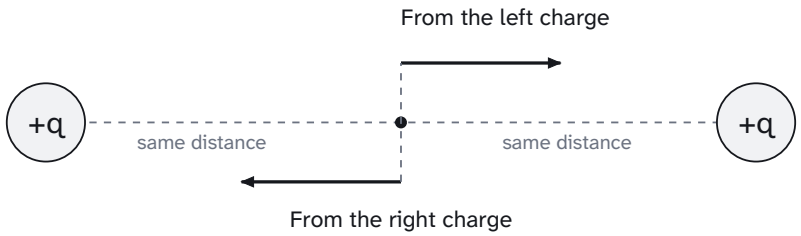
EM

Electricity and Magnetism

Understand the idea. Build the method. Earn the marks.

FIG. 02 / ELECTRICITY

Equal source fields cancel at the midpoint



Net field = 0

Takeaway: equal field magnitudes cancel only when their directions oppose.

Document	Electricity and Magnetism
Level / lessons	3 / 12
Edition	November 2026 / Bench Sheet
Pages	67

Start here

Begin with one lesson and a pencil. Follow the stages below, then practise without notes.

1. Read the Idea and describe it in your own words before looking at the rule.
2. Check what every symbol means and when the rule applies.
3. Cover the worked steps. Predict the next step, then compare.
4. Use the working grid to attempt Your turn. Check the answers in the lesson review section.
5. Attempt the subject tests without notes. Use the explanations to identify the exact step you missed.

Read the page at a glance

Rule

$$v = \frac{d}{t}$$

A framed rule is something to select and apply, not just memorise. Here, average speed is distance divided by elapsed time.

Your turn

Blue marks a task for you. Write a prediction or calculation before checking.

Watch out

An orange note marks a common error. Example: do not confuse distance with displacement.

Before you begin

- Have a calculator, pencil and the units used in your course.
- Review basic rearrangement and graph reading when a lesson lists them as pre-requisites.
- Use the formula sheet and symbol glossary for revision; the same letter can have different meanings in different lessons.

Use the linked contents and alphabetical topic index to find a lesson. Marking points are study guidance, not an official marking scheme.

Your study route

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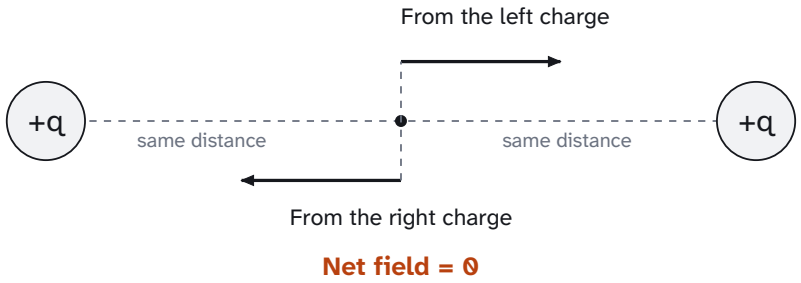
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Topic index **67**

Electricity and Magnetism

FIG. 02 / ELECTRICITY

Equal source fields cancel at the midpoint



Takeaway: equal field magnitudes cancel only when their directions oppose.

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

Subject map

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

Follow this reading route from foundations to applications. Each line links to its lesson; revisit the stated prerequisites whenever a step feels unfamiliar.

01 Charges, conductors and Coulomb force p.6

02 Electric field and superposition p.9

03 Flux and Gauss law p.13

04 Symmetry and conductors p.16

05 Potential is not field p.19

06 Capacitors and dielectrics p.22

07 Current, resistance and electrical power p.26

08 DC networks, Kirchhoff rules and RC behaviour p.29

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11 Induction and Lenz law p.40

12 Inductance, AC and electromagnetic waves p.43

Charges, conductors and Coulomb force

EM / 01

- Predict attraction or repulsion from charge signs.
- Use simple Coulomb-force ratios without unnecessary arithmetic.

You need: Positive and negative charge; squares; micro means one millionth.

01 Idea

Charge is conserved and occurs in integer multiples of the elementary charge. Ordinary charging transfers electrons. Like charges repel and unlike charges attract. Conductors let charge move readily; insulators restrict it. A neutral object can still be attracted by polarization.

Coulomb-force ratios let you compare two situations without knowing every charge value. Imagine moving the same two charges farther apart: the effect spreads according to the square of separation. Charge signs tell attraction or repulsion, while charge sizes and separation set the magnitude.

- Doubling separation squares the distance factor, making the force one quarter as large.
- Changing only one charge magnitude scales the force directly without changing the separation factor.

02 Rule and symbols

Rule

$$F = k|q_1 q_2|/r^2$$

$$k = 8.99 \times 10^9 \ \mathrm{N\,m^2/C^2}$$

F electrostatic force magnitude / N

q_1, q_2 source charges / C

r separation of charges / m

Use this when:

Coulomb calculations here use point charges, or a stated model equivalent to point charges.

03 Method

1. Mark the signs and positions of both charges.
2. Predict attraction or repulsion before calculating magnitude.
3. For a ratio, compare the charge product and the square of the separation.
4. For a numerical force, convert charges and lengths to SI units first.

Use a ratio when a force is already known and only a charge or separation changes.

04 Worked example

Two charges at distance r exert force F.

01 Keep both charges fixed and replace the separation r by 2r.

02 The denominator becomes $(2r)^2 = 4r^2$, so the force becomes F/4.

03 If instead one charge doubles at the original separation, its product with the other charge doubles, so force becomes 2F.

04 Changing separation or charge magnitude alone does not change attraction into repulsion.

Watch out

A negative charge does not mean a negative force magnitude.

Exam habit

A magnitude answer is positive; state attraction or repulsion separately to communicate direction.

05 Your turn

Two charges exert 12 N at separation r. What is the force at separation 2r, and at r if one charge is tripled?

If you get stuck, uncover one hint at a time:

1. Treat the two requested situations separately, each starting from the original 12 N force.
2. For doubled separation, square the separation-change factor before applying it to the force.
3. For one charge tripled at the original separation, apply only the charge-change factor. State both results with newtons.

06 Check your understanding

Tripling separation with fixed charges changes the force by what factor?

Two unlike charges are moved farther apart without changing their signs. What changes?

1. The attraction becomes weaker
2. Attraction becomes repulsion
3. The force keeps the same magnitude

Answers and explanations: page 46.

Electric field and superposition

EM / 02

- Combine electric fields by their directions at an observation point.
- Explain why a negative test charge feels force opposite the field.

You need: Vector addition; positive and negative charges; multiplication.

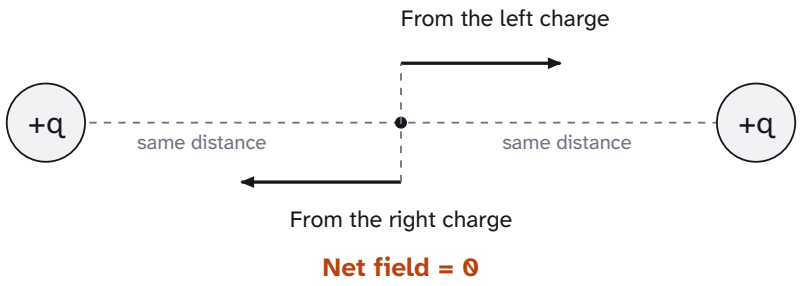
01 Idea

The electric field gives force per unit positive test charge. Field arrows point away from positive sources and toward negative sources. The field exists whether or not a test charge is present. Add fields as vectors, so equal magnitudes can cancel.

A field is a direction-and-strength instruction already present at a location. Its arrow says how a positive test charge would be pushed. An actual negative charge reads the same instruction with the direction reversed, so finding a field and finding a charge force are separate steps.

FIG. 02 / ELECTRICITY

Equal source fields cancel at the midpoint



Takeaway: equal field magnitudes cancel only when their directions oppose.

Figure 02. Equal positive charges give zero net field at the midpoint, so the force on a test charge there is zero. The two field arrows are offset vertically only for clarity; both represent fields at the midpoint.

- Multiplying by a negative charge reverses direction but does not make the force magnitude negative.
- The field is defined by the source arrangement and does not require the test charge to be present.

02 Rule and symbols

Rule

$$E = F/q_{\text{test}}$$
$$E = k|Q|/r^2$$
$$\vec{F} = q\vec{E}$$

E electric field magnitude / N/C

q charge experiencing the force / C

F force magnitude / N

03 Method

1. Choose the observation point and draw the field arrow from each source there.
2. Point away from a positive source and toward a negative source.
3. Add signed collinear fields or use vector components when needed.
4. Only after finding the net field, use the test-charge sign to obtain force direction.

Separate field direction from force direction whenever the question supplies a negative charge.

04 Worked example

At the midpoint between equal positive charges, the two electric fields oppose and cancel.

01 At the midpoint between equal positive charges, both source distances are the same, so field magnitudes match.

02 The left positive charge points the field right; the right positive charge points it left. The net field is zero.

03 Replace the right source by an equal negative charge: its field now points toward it, also to the right.

04 The two fields therefore reinforce, pointing from the positive source toward the negative source.

Watch out

Zero net field at a point does not mean each source field is zero.

Exam habit

Draw field arrows before multiplying by a test charge; otherwise a negative test

charge can lead to two direction reversals.

05 Your turn

A field of 200 N/C points east. Find the force on a charge of -0.003 C.

If you get stuck, uncover one hint at a time:

1. The field direction is east. Decide the force direction using the sign of the supplied charge before calculating.
2. Multiply the field magnitude 200 N/C by the charge magnitude 0.003 C.
3. State the positive force magnitude in newtons and its compass direction. The minus sign belongs to the directional reasoning, not a magnitude.

06 Check your understanding

If the net field is zero at a point, is a charge placed there forced in either direction by that field?

Two small test charges, one positive and one negative with equal magnitudes, are placed separately at the same point in a nonzero electric field. Their forces have:

1. Different magnitudes solely because their signs differ
2. Equal magnitudes and opposite directions
3. Equal magnitudes and the same direction

Answers and explanations: page 46.

Flux and Gauss law

EM / 03

- Use the surface normal to determine electric flux.
- Distinguish enclosed charge from outside charge in Gauss law.

You need: Surface area; normal means perpendicular; cosine at 0° and 90°.

01 Idea

Electric flux measures the field passing through an oriented surface. The angle is measured from the surface normal, not the surface itself. Net flux through a closed surface depends on enclosed charge. An outside charge can affect the field at the surface but contributes zero net closed-surface flux.

Flux counts the field crossing a surface rather than the field lying along it. Imagine rain falling through a hoop: turning the hoop changes how much rain crosses its opening even when the rain itself stays the same. A surface normal records that orientation.

- A field parallel to the normal crosses the full effective area.
- A field tangent to the surface has no crossing component, even if its magnitude is large.

02 Rule and symbols

Rule

$$\Phi = EA\cos\theta$$

$$\Phi_{\text{closed}} = Q_{\text{enclosed}}/\epsilon_0$$

Φ electric flux / N m^2/C

E electric field magnitude / N/C

A surface area / m^2

Q_{enclosed} net charge inside a closed surface / C

03 Method

1. Decide whether the question concerns one flat surface or a complete closed surface.
2. For a flat surface, draw the normal and identify its angle to the field.
3. For a closed surface, add only the enclosed charges with signs.
4. Interpret positive closed flux as net outward flux and negative as net inward flux.

Use flat-surface flux when area, a uniform field and orientation are given; use enclosed charge for net closed-surface flux.

04 Worked example

A cube enclosing +q has net outward flux $\frac{q}{\varepsilon_0}$.

01 The cube is a closed surface enclosing net charge +q.

02 Gauss law fixes its total outward flux as q divided by ϵ0.

03 Moving q off-centre leaves enclosed charge unchanged, so total flux is unchanged.

04 The nearby faces can now have stronger local fields; Gauss law does not say each face receives an equal share after the move.

Watch out

Zero net closed-surface flux does not imply zero field everywhere.

Exam habit

The angle in EA cos θ is to the normal; a field parallel to the surface gives zero flux.

05 Your turn

A uniform field of 50 N/C passes through a flat area of 0.20 square metres. The chosen surface normal points in the same direction as the field. Find flux. What if the normal is perpendicular to the field?

If you get stuck, uncover one hint at a time:

1. Draw the chosen normal as an arrow. In the first case it points with the field, so identify their angle.
2. Multiply field 50 N/C by area 0.20 m^2 and the cosine of that normal angle.
3. For the second case change only the cosine factor. Keep flux units distinct from field units.

06 Check your understanding

Can an external charge produce a nonzero field on a box while contributing zero net flux through it?

A flat surface rotates from normal parallel to the field to field tangent to the surface. What happens to flux magnitude?

1. It stays constant because field strength is unchanged
2. It increases because the surface turns
3. It falls from its maximum to zero

Answers and explanations: page 46.

Symmetry and conductors

EM / 04

- State the electrostatic properties of conducting material.
- Use the inverse-square field rule outside a spherical conductor.

You need: Electric field; conductor versus insulator; inverse-square ratios.

01 Idea

Gauss law is always valid, but it easily gives field magnitude only when symmetry makes the surface calculation simple. In electrostatic equilibrium the field inside conducting material is zero and excess charge lies on its surface. A conductor is an equipotential. For the outside-sphere inverse-square rule, assume an isolated conducting sphere with no external fields.

On an isolated conducting sphere, charge can rearrange until the conductor is in electrostatic equilibrium. Outside this symmetric arrangement, its field follows the same distance pattern as one central point charge. The relevant distance therefore starts at the centre, not at the surface.

- Spherical symmetry makes all outside points at the same centre distance equivalent.
- The inverse-square outside field weakens by a factor of four when centre distance doubles.

02 Rule and symbols

Rule

$$E_{\text{outside sphere}} = k|Q|/r^2$$
$$(r > R)$$

Q net sphere charge / C

R sphere radius / m

r distance from sphere centre / m

E electric field magnitude / N/C

Use this when:

The sphere is isolated with spherical symmetry and is in electrostatic equilibrium.

03 Method

1. Check that the conductor is in electrostatic equilibrium.
2. Distinguish a point in the conducting material from a point outside the sphere.
3. For an outside point measure distance from the centre, not from the surface.
4. Use the point-charge field or a simple ratio, then describe the radial direction.

Use the outside-sphere comparison only when the sphere is isolated as stated and both points lie outside it.

04 Worked example

A charged conducting sphere behaves outside like a point charge at its centre.

01 Outside a spherically symmetric charged conducting sphere, the field has the same form as a charge at its centre.

02 Compare two outside distances r and 2r.

03 The square of the distance grows by a factor of four, so the field magnitude becomes E/4.

04 This comparison uses centre distances; doubling the gap from the surface would generally give a different ratio.

Watch out

Do not extend the zero-field-in-metal statement to every cavity containing a charge.

Exam habit

Write r > R beside the outside-sphere formula to show that its domain has been checked.

05 Your turn

An isolated charged conducting sphere, with no external fields, has field magnitude 80 N/C at r = 0.20 m. Find the field at r = 0.40 m. Both distances are from the centre and outside the sphere.

If you get stuck, uncover one hint at a time:

1. Check that both stated distances start at the centre and exceed the sphere radius as specified.
2. Compare 0.40 m with 0.20 m and square that distance ratio.
3. Divide the original 80 N/C by the squared distance factor. No unknown total charge is needed.

06 Check your understanding

A field is 80 N/C at 0.20 m from the centre. What is it at 0.40 m if both points are outside?

Why can doubling the gap above a sphere surface differ from doubling the distance used in its field formula?

1. The formula measures distance from the centre
2. The formula always uses distance from the nearest charge on the surface
3. Outside fields never depend on distance

Answers and explanations: page 46.

Potential is not field

EM / 05

- Add scalar potentials with their charge signs.
- Relate potential difference to energy change and field work.

You need: Scalar versus vector; signed energy changes; potential energy.

01 Idea

Electric potential is energy per unit charge, a scalar. Electric field is a vector. Add potentials with charge signs, but add fields with directions. Work by the electric field equals the decrease in electric potential energy.

Potential is a location-based energy-per-charge value, rather like height in a gravitational comparison. A change in potential tells how the potential energy of a particular charge changes. The electric field work has the opposite sign because energy given to kinetic motion comes from a decrease in potential energy.

- A positive charge moving to higher potential gains electric potential energy.
- Positive potential-energy gain means the electric field does negative work over that move.

02 Rule and symbols

Rule

$$V = kQ/r \text{ (zero at infinity)}$$

$$\Delta U = q\Delta V$$

$$W_{\text{field}} = -\Delta U$$

V electric potential / V = J/C

ΔV final minus initial potential / V

ΔU change in potential energy / J

q moving charge / C

03 Method

1. Separate a field question, requiring direction, from a potential question, requiring signed scalar addition.
2. Choose the potential reference supplied by the problem.
3. Calculate energy change from charge times final-minus-initial potential.
4. Reverse the sign of that energy change to obtain work by the electric field.

When a question gives a charge and a potential difference, use the energy change directly rather than inventing a field or travel distance.

04 Worked example

Equal positive charges can give zero field at their midpoint but positive potential there.

01 At the midpoint of two equal positive charges, equal field arrows point in opposite directions and cancel.

02 Each positive charge contributes a positive potential at that point when zero potential is chosen at infinity.

03 The two potentials add rather than cancel because potential has no directional arrow.

04 Thus zero net field and positive potential can occur at the same point without contradiction.

Watch out

A charge moving to lower potential does not always lose potential energy; a negative charge does the opposite.

Exam habit

Write ΔV = Vfinal - Vinitial before calculating; swapping the endpoints reverses both energy and work signs.

05 Your turn

A +2 C charge moves through a potential difference of +3 V. Find its potential-energy change and the work done by the field.

If you get stuck, uncover one hint at a time:

1. Treat the stated +3 V as final potential minus initial potential. Identify the moving charge sign.
2. Multiply charge +2 C by the potential difference to obtain the signed potential-energy change.
3. Reverse that energy-change sign for work done by the field. Report each quantity in joules and label them separately.

06 Check your understanding

A +2 C charge moves through $\Delta V = +3$ V. What are ΔU and work by the field?

A negative charge moves through a positive potential difference. What is the sign of its potential-energy change?

1. Always zero
2. Negative
3. Positive

Answers and explanations: page 46.

Capacitors and dielectrics

EM / 06

- Distinguish series and parallel capacitor connections.
- Calculate stored charge and identify the effect of a dielectric.

You need: Potential difference; charge; reciprocals and addition.

01 Idea

A capacitor stores separated charge and electric-field energy. Capacitance describes charge stored per volt. Parallel capacitors share voltage and their capacitances add. Series capacitors carry equal charge magnitude and have a smaller equivalent capacitance.

Capacitance describes how much separated charge corresponds to each volt across two plates. A capacitor with a larger capacitance can store more charge at the same voltage. It is the connection to a battery or isolation of the capacitor that decides which quantity stays fixed during a change.

- At fixed voltage, increasing capacitance increases stored charge.
- A microfarad multiplied by a volt gives a microcoulomb, so compatible prefixes can simplify a calculation.

02 Rule and symbols

Rule

$$Q = CV$$

$$U = \tfrac{1}{2}CV^2$$

$$C_{\parallel} = C_1 + C_2$$

$$1/C_s = 1/C_1 + 1/C_2$$

C capacitance / F

Q stored charge magnitude on either plate / C

V potential difference between plates / V

U stored electric energy / J

03 Method

1. Identify shared nodes: parallel capacitors share voltage, while ideal series capacitors carry equal charge magnitudes.
2. Add capacitances for parallel or add their reciprocals for series.
3. Use Q = CV with compatible prefixes such as μF times V giving μC.
4. For dielectric questions first check whether voltage or charge is held fixed.

Use the charge-capacitance-voltage relation when any two of those three quantities are supplied.

04 Worked example

Two 6 μF capacitors give 12 μF in parallel and 3 μF in series.

01 For two 6 μF capacitors in parallel, capacitances add: 6 + 6 = 12 μF.

02 For series, 1/Cs = 1/6 + 1/6 = 1/3 in reciprocal μF, so Cs = 3 μF.

03 The series result is smaller than either individual capacitor, providing a quick plausibility check.

04 Inserting a suitable dielectric increases capacitance; the resulting charge or voltage change depends on the connection to a battery.

Watch out

Do not use resistor series/parallel rules for capacitors.

Exam habit

For series, invert the reciprocal sum at the end; stopping at 1/C is a common lost result mark.

05 Your turn

A 4 μF capacitor is connected to 12 V. Find the charge in microcoulombs.

If you get stuck, uncover one hint at a time:

1. The capacitor is connected across 12 V. Identify whether charge follows by multiplying or dividing capacitance and voltage.
2. Use capacitance 4 microfarads and the supplied voltage.
3. A result from microfarads times volts is in microcoulombs. If using farads instead, convert the final coulombs back consistently.

06 Check your understanding

What charge magnitude is stored by 4 μF at 12 V?

A capacitor stays connected to an ideal battery while its capacitance increases. What happens?

1. Charge stays fixed and voltage increases
2. Both voltage and charge must stay fixed
3. Voltage stays fixed and stored charge increases

Answers and explanations: page 46.

What changes a capacitor’s capacitance?

These simple comparisons cover the geometry and dielectric ideas in the lecture without a derivation.

- Use the first formula for parallel plates with a small gap compared with plate size, so edge effects can be ignored.
- Here A is overlapping plate area, d is separation, and ε describes the material filling the gap.
- Doubling area doubles capacitance. Doubling separation halves it.

- A dielectric filling the gap increases capacitance. Battery attached: voltage stays fixed. Disconnected and isolated: charge stays fixed.
- $C = \dfrac{\varepsilon A}{d}$
- $Q = CV$

An isolated charged capacitor has capacitance $2\,\mu\mathrm{F}$ and voltage $12\,\mathrm{V}$. A dielectric triples its capacitance. Find the new charge and voltage.

1. Initial charge: $Q = CV = 2 \times 12 = 24\,\mu\mathrm{C}$.
2. Isolation keeps the charge at $24\,\mu\mathrm{C}$.
3. New capacitance: $6\,\mu\mathrm{F}$; new voltage: $V = 24/6 = 4\,\mathrm{V}$.

Charge stays $24\,\mu\mathrm{C}$; voltage falls to $4\,\mathrm{V}$.

Current, resistance and electrical power

EM / 07

- Find current, voltage and power for an ohmic resistor.
- Predict how wire length and area change resistance.

You need: Charge flow; multiplication and division; circuit voltage.

01 Idea

Current is the rate of charge flow. Conventional current follows positive-charge flow, opposite electron drift in a metal. Resistance depends on material and geometry. Ohm law describes an ohmic element under fixed physical conditions.

Voltage is the energy transferred per unit charge, while current counts charge passing each second. Multiplying them gives energy per second, which is power. Resistance connects the current through a resistor with the voltage across that same resistor; those locations matter in a circuit.

- For a given resistor current, the voltage drop follows its resistance.
- Power follows both the amount of charge passing each second and the energy transferred by each unit charge.

02 Rule and symbols

Rule

$$I = \Delta Q/\Delta t$$

$$V = IR$$

$$R = \rho L/A$$

$$P = VI$$

I conventional current / A

R resistance / Ω

V potential difference / V

P electrical power / W

ρ resistivity / Ω m

Use this when:

Ohm law is used for an ohmic resistor under unchanged physical conditions.

03 Method

1. Label voltage across the resistor and current through it.
2. Use V = IR to find the missing electrical quantity.
3. Calculate power using VI or an equivalent resistor relation.
4. For wire comparisons hold material and temperature fixed before changing length or area.

When current and resistance are supplied, find the resistor voltage first, then use voltage times current for power.

04 Worked example

A 6 Ω resistor across 12 V carries 2 A and dissipates 24 W.

01 The 6 Ω resistor has 12 V across it, so I = 12/6 = 2 A.

02 Its electrical power is VI = 12 times 2 = 24 W.

03 Power of 24 W means 24 joules of energy are transferred each second.

04 A longer wire with unchanged material, area and temperature has larger resistance because R is proportional to length.

Watch out

More resistance lowers power at fixed voltage but raises power at fixed current.

Exam habit

Do not confuse power in watts with energy in joules; multiply power by time when energy is requested.

05 Your turn

A 3 Ω resistor carries 2 A. Find voltage and power.

If you get stuck, uncover one hint at a time:

1. Mark the given 2 A as current through the 3-ohm resistor. Which relation gives its voltage drop?
2. Multiply current by resistance to obtain voltage.
3. Use that voltage with the same current to obtain power. Check that the units are volts for the first result and watts for the second.

06 Check your understanding

What happens to resistance if a wire length doubles but material, area and temperature stay fixed?

The current through the same fixed resistor doubles. What happens to its power?

1. It becomes four times as large
2. It doubles
3. It stays fixed because resistance stays fixed

Answers and explanations: page 46.

DC networks, Kirchhoff rules and RC behaviour

EM / 08

- Reduce simple resistor networks and recover branch currents.
- Describe the initial and final behaviour of an RC charging circuit.

You need: Ohm law; series and parallel connections; charge and energy conservation.

01 Idea

Series resistors share current; parallel resistors share voltage. Kirchhoff junction rules express charge conservation, and loop rules express energy conservation. An initially uncharged capacitor charges gradually through a resistor and eventually blocks steady DC in the ideal model.

In a series path, charge has no branch in which to split, so the same current passes through each resistor. The source voltage is shared as energy drops along the path. A larger resistance takes a larger voltage share because the current is common.

- Series resistances add because their voltage drops add at the same current.
- Checking that the individual drops sum to the source voltage tests the energy bookkeeping.

02 Rule and symbols

Rule

$$R_s = R_1 + R_2$$

$$1/R_p = 1/R_1 + 1/R_2$$

$$\tau = RC$$

R_{eq} equivalent resistance / Ω

I current / A

V voltage / V

τ RC time constant / s

03 Method

1. Identify series elements by shared current and parallel elements by shared voltage.
2. Reduce the network and calculate total source current.
3. Return to the original circuit to calculate branch currents or voltage drops.
4. Check junction current sums and loop voltage sums; for RC questions distinguish transient from long-time behaviour.

Use series reduction when every charge passing one element must pass the next without a junction.

04 Worked example

Two 6 Ω resistors in parallel give 3 Ω.

01 Both 6 Ω resistors are connected across the same two nodes, so they are parallel.

02 Their equal-resistor equivalent is 6/2 = 3 Ω.

03 The 12 V source therefore supplies 12/3 = 4 A total.

04 Each branch still has 12 V and carries 12/6 = 2 A; the branch currents sum to the 4 A source current.

Watch out

Adjacent components in a drawing are not necessarily in series.

Exam habit

After finding an equivalent resistance, redraw or relabel the original circuit before assigning individual voltage drops.

05 Your turn

A 4 Ω and 8 Ω resistor are in series across 24 V. Find total current and each voltage drop.

If you get stuck, uncover one hint at a time:

1. Confirm there is one series path. Add the two resistances before finding source current.
2. Divide the 24 V source voltage by the total resistance.
3. Use that same current with each resistance separately. Verify that the two voltage drops add to 24 V.

06 Check your understanding

After a long time, does an ideal capacitor in a DC series RC circuit carry a steady current?

A 4-ohm resistor and an 8-ohm resistor carry the same series current. Which has the larger voltage drop?

1. The drops must be equal because current is equal
2. The 8-ohm resistor
3. The 4-ohm resistor

Answers and explanations: page 46.

One time constant: the 63% and 37% rules

Recognize how quickly a capacitor charges or discharges without deriving an exponential.

- Use an ideal series resistor-capacitor circuit with constant R and C. The time constant τ is measured in seconds.
- Charging from zero: after one time constant, capacitor voltage and charge reach about 63% of their final values.
- Discharging: after one time constant, about 37% of the starting voltage and charge remain.
- A larger resistance or capacitance makes the change slower.
- $\tau = RC$
- $1 - e^{-1} \approx 0.632$
- $e^{-1} \approx 0.368$

A capacitor charges from zero toward $10\,\mathrm{V}$. Its time constant is $2\,\mathrm{s}$. Estimate its voltage after $2\,\mathrm{s}$.

1. The elapsed time is one time constant.
2. Multiply final voltage by 0.632: $10 \times 0.632 = 6.32\,\mathrm{V}$.

About $6.3\,\mathrm{V}$. It is not fully charged after one time constant.

Kirchhoff signs and a real battery

Choosing consistent signs is the main skill. A negative current answer tells you the actual direction.

- At a junction, charge does not keep building up: add entering and leaving currents separately.
- Choose a loop direction. Across a resistor, travelling with the assumed current gives $-IR$; travelling against it gives $+IR$.
- Across a battery, going from its negative to positive terminal gives $+\mathcal{E}$; the reverse gives $-\mathcal{E}$.
- For a battery supplying current, internal resistance r causes the terminal voltage to be less than its emf $\mathcal{E}$.
- current in $=$ current out
- $\sum \Delta V = 0$

- $V_{\text{terminal}} = \mathcal{E} - Ir$ (battery delivering current)

A battery has emf $12\,\text{V}$ and internal resistance $1\,\Omega$. It supplies a $5\,\Omega$ resistor. Find current and terminal voltage.

1. Use one loop: $12 - I(1) - I(5) = 0$.
2. Total resistance is $6\,\Omega$, so $I = 12/6 = 2\,\text{A}$.
3. Terminal voltage is $12 - (2)(1) = 10\,\text{V}$. Check the external resistor: $IR = (2)(5) = 10\,\text{V}$.

Current $2\,\text{A}$; terminal voltage $10\,\text{V}$.

Magnetic force: direction before calculation

EM / 09

- Determine when magnetic force vanishes or is largest.
- Explain circular motion without a change in kinetic energy.

You need: Perpendicular directions; circular acceleration; charge sign.

01 Idea

A magnetic field exerts force on a moving charge when its motion has a perpendicular component. The force is perpendicular to velocity and therefore does no work on a point charge. It can change direction without changing speed.

Magnetic force acts sideways relative to moving charge. For a straight current-carrying wire, the current direction plays the corresponding role. Geometry matters: a wire across the field has maximum magnetic force, whereas a wire along the field has no perpendicular component and no magnetic force.

- A perpendicular arrangement uses the full field-current-length product because the sine factor is one.
- Reversing current reverses force direction but does not change magnitude when other factors stay fixed.

02 Rule and symbols

Rule

$$F = |q|vB\sin\theta$$

$$F_{\text{wire}} = BIL\sin\theta$$

q particle charge / C

v particle speed / m/s

B magnetic field magnitude / T

F magnetic force magnitude / N

θ angle between velocity and field / degrees or radians

03 Method

1. Decide whether the object is a moving charge or a current-carrying wire.
2. Identify the angle to the magnetic field and the perpendicular motion or wire length.
3. Use the right-hand rule for positive charge or conventional current; reverse for negative charge.
4. Calculate magnitude and check that the force is perpendicular to motion.

Use the wire-force relation when current, straight wire length and magnetic field are given; do not substitute particle speed.

04 Worked example

A positive charge moving perpendicular to a uniform magnetic field follows a circular path if no other force acts.

01 For the positive charge in the example, velocity is perpendicular to the uniform field, so the magnetic force magnitude is qvB.

02 At each point the force is perpendicular to velocity and turns the particle inward.

03 Because the force does no work, speed and kinetic energy remain constant.

04 With no other forces, the continual turning gives a circular path in the plane perpendicular to the field.

Watch out

Do not apply the right-hand direction unchanged to an electron.

Exam habit

A direction answer needs the charge sign; a right-hand-rule result for a positive charge must be reversed for an electron.

05 Your turn

A straight 0.50 m wire carries 2 A perpendicular to a 0.30 T field. Find force magnitude.

If you get stuck, uncover one hint at a time:

1. This is a current-carrying wire, not a single moving particle. Choose the relation involving field, current and length.
2. The wire is perpendicular to the field, so the angle factor is one.
3. Multiply 0.30 T, 2 A and 0.50 m. The question asks for magnitude, so no charge sign is needed.

06 Check your understanding

Does a charge moving exactly along a magnetic field feel magnetic force?

A straight wire initially lies perpendicular to a uniform field. It rotates until parallel while current stays fixed. Its magnetic force magnitude:

1. Stays unchanged
2. Doubles
3. Falls to zero

Answers and explanations: page 46.

Sources of magnetic fields

EM / 10

- Relate a current to the direction of the magnetic field it produces.
- Use simple distance and current ratios for a long wire.

You need: Current; inverse proportionality; circular field-line diagrams.

01 Idea

Currents produce magnetic fields. Around a long straight wire the field forms circles. Inside an ideal long solenoid the field is approximately uniform. For two parallel wires, currents in the same direction attract and opposite currents repel.

A straight current-carrying wire has magnetic field lines that circle the wire. Moving farther away weakens the field, but its distance pattern differs from the field of a point charge. Recognizing the source geometry prevents using the familiar but incorrect inverse-square rule here.

- For an ideal long wire, magnetic field magnitude is inversely proportional to perpendicular distance.
- At unchanged current, a threefold increase in distance therefore gives a threefold decrease rather than a ninefold decrease.

02 Rule and symbols

Rule

$$B_{\text{wire}} = \mu_0 I/(2\pi r)$$

$$B_{\text{solenoid}} = \mu_0 n I$$

B magnetic field magnitude / T

I wire current / A

r perpendicular distance from wire / m

n solenoid turns per unit length / m^{-1}

Use this when:

The wire is sufficiently long and the observation point sufficiently far from its ends for the long-wire approximation.

03 Method

1. Identify whether the source is a straight wire, a solenoid or a pair of wires.
2. For a straight wire, use the right-hand grip rule to draw field circles.
3. Apply the matching field relation; for ratios hold unstated factors fixed.
4. For paired wires determine attraction or repulsion from relative current directions.

The phrase long straight wire signals the current-over-distance rule, not the point-charge inverse-square rule.

04 Worked example

Doubling distance from a long wire halves its magnetic field; doubling current doubles the field.

01 For the long wire, B is proportional to I/r.

02 Moving from r to 2r at the same current gives half the original field.

03 Doubling current at fixed r instead doubles the field.

04 If current and distance both double, the two factors cancel and field magnitude stays the same.

Watch out

The wire field varies as inverse distance, not inverse distance squared.

Exam habit

Use 1/r for the long-wire field; the $1/r^2$ rule belongs to a different source geometry.

05 Your turn

The field is 30 microtesla at 2 cm from a long straight wire. Find the field at 6 cm at the same current.

If you get stuck, uncover one hint at a time:

1. Keep the current fixed as stated. Compare the new perpendicular distance with the original distance.
2. Use an inverse-distance ratio for a long wire; do not square the ratio.
3. Apply that factor to 30 microtesla. Centimetres can remain in the ratio because both distances use the same unit.

06 Check your understanding

A wire gives 30 μT at 2 cm. What field does it give at 6 cm with the same current?

Current and perpendicular distance from a long wire both double. What happens to magnetic field magnitude?

1. It stays unchanged
2. It doubles
3. It halves

Answers and explanations: page 46.

Induction and Lenz law

EM / 11

- Identify which change produces an induced emf.
- Use Lenz law to choose the induced current direction.

You need: Magnetic field direction; loop area; clockwise versus counterclockwise viewed from a stated side.

01 Idea

An induced emf requires a change in magnetic flux, which can arise from changing field, area or orientation. Lenz law sets the direction: the induced effect opposes the change of flux. A static magnet near a stationary loop does not produce a sustained induced current.

Induction responds to a changing magnetic flux through a loop, rather than merely to the presence of a magnet. A faster change gives a larger average emf. A coil repeats the same flux-change effect through many turns, so their induced emfs add when the turns are linked alike.

- More turns multiply the emf when each turn experiences the same flux change.
- Spreading the same flux change over a longer time reduces the average emf.

02 Rule and symbols

Rule

$$|\mathcal{E}| = N|\Delta\Phi_B|/\Delta t$$

$$\Phi_B = BA\cos\theta$$

Φ_B magnetic flux through one turn / Wb

N number of coil turns / dimensionless

$\mathcal{E}$ induced emf / V

Δt time interval of flux change / s

03 Method

1. State the original flux direction and whether its magnitude increases or decreases.
2. Choose an induced field that opposes that change.
3. Use the grip rule to convert the required field to current direction, naming the viewing side.
4. For magnitude, multiply the flux change per turn by the number of turns and divide by elapsed time.

Look for a change in field, loop area or loop orientation over a stated time; identify whether flux is per turn.

04 Worked example

If outward flux increases, the induced field points inward.

01 In the example, outward flux through the loop is increasing.

02 The induced effect must oppose the increase, so its own field points inward.

03 Viewed from the side toward which the original flux points outward, the inward field requires clockwise current.

04 If the outward flux were decreasing instead, the induced field and current direction would both reverse.

Watch out

Opposes the change is not the same as always opposes the original field.

Exam habit

Say oppose the change in flux, not oppose the flux: a decreasing field produces a reinforcing induced field.

05 Your turn

A 20-turn coil has flux per turn changing by 0.03 Wb in 0.10 s. Find average emf magnitude.

If you get stuck, uncover one hint at a time:

1. The problem gives flux change per turn. Decide where the 20 turns enter the total effect.
2. Multiply turn count by flux change per turn, then divide by the 0.10 s interval.
3. Use the magnitude of the change because magnitude is requested. A current direction would require separate Lenz-law information.

06 Check your understanding

Does a stationary loop near an unchanging stationary magnet have a sustained induced emf?

The same flux change per turn occurs in half the time. With turn count unchanged, average emf magnitude:

1. Stays unchanged
2. Doubles
3. Halves

Answers and explanations: page 46.

Inductance, AC and electromagnetic waves

EM / 12

- Convert sinusoidal peak voltage to RMS and calculate resistive heating.
- Relate electromagnetic frequency to vacuum wavelength.

You need: Square roots; frequency and wavelength; electrical power.

01 Idea

Course-plan extension: an inductor resists changes of current and stores magnetic energy. AC reverses direction periodically. RMS values describe equivalent heating for resistive loads. In vacuum an EM wave has mutually perpendicular electric field, magnetic field and travel direction.

An alternating voltage has a changing instantaneous value, so its peak is not its steady heating equivalent. RMS voltage compresses that varying waveform into the DC voltage that would produce the same average heating in a resistor. The familiar peak conversion depends on the waveform being sinusoidal.

- Resistive heating depends on voltage squared, so simply averaging positive and negative voltage would miss the heating.
- A sinusoid has RMS voltage below its peak because it spends only part of each cycle near the peak.

02 Rule and symbols

Rule

$$U_L = \tfrac{1}{2} L I^2$$

$$V_{\mathrm{rms}} = V_{\mathrm{peak}} / \sqrt{2} \text{ (sinusoidal)}$$

$$c = f\lambda \text{ (vacuum)}$$

V_{peak} maximum sinusoidal voltage magnitude / V

V_{rms} root-mean-square voltage / V

L inductance / H

f wave frequency / Hz

λ wavelength / m

Use this when:

The peak-to-RMS conversion assumes a sinusoid; the wavelength calculation assumes propagation in vacuum.

03 Method

1. Identify whether the question concerns an inductor, sinusoidal AC or an electromagnetic wave.
2. For sinusoidal AC divide peak voltage by the square root of two.
3. Use RMS values for average power in a pure resistor.
4. For vacuum waves calculate wavelength as c divided by frequency and convert metres to the requested unit.

Use RMS values for average resistive power; use the peak-to-RMS conversion only for a specified sinusoid.

04 Worked example

A sinusoidal supply with peak voltage 141.4 V has RMS voltage 100 V.

01 For the stated sinusoidal supply, 141.4 divided by $\sqrt{2}$ is approximately 100 V RMS.

02 This RMS value matches the heating effect of 100 V DC in the same pure resistor.

03 For the separate light example, wavelength is 3.00×10^{8} divided by 6×10^{14} = 5×10^{-7} m.

04 Since one nanometre is 10^{-9} m, the wavelength is 500 nm.

Watch out

An ideal transformer does not continuously transform steady DC after switching transients.

Exam habit

Keep this extension conceptual: use RMS for resistor power and $c = f\lambda$ for waves; do not mix peak and RMS values.

05 Your turn

A sinusoidal voltage has peak value 20 V. Find its RMS value, then the average power in a 10 Ω resistor.

If you get stuck, uncover one hint at a time:

1. Separate the request into RMS voltage and average resistor power. The supplied 20 V is a peak value.
2. Divide the peak by the square root of two to obtain the sinusoidal RMS value.
3. Square that RMS voltage and divide by 10 ohms. Keep a few extra digits until the final rounding.

06 Check your understanding

A 20 V peak sinusoid is applied to 10 Ω. What is the average power?

A symmetric sinusoidal voltage has zero average voltage over a cycle. Does a resistor receive zero average power?

1. Yes; positive and negative heating cancel
2. Only the positive half-cycle heats it
3. No; both voltage polarities produce heating

Answers and explanations: page 46.

Lesson review and answers

Charges, conductors and Coulomb force

Return to lesson: page 6.

Your turn: It becomes one ninth as large.

Concept check: 1. Correct. Opposite signs still attract, but larger separation reduces magnitude.

Written method: At 2r: 12/4 = 3 N. Tripling one charge at r: 36 N. Two marks for each ratio and result.

Electric field and superposition

Return to lesson: page 9.

Your turn: No. The electric force qE is zero there in the stated source-field model.

Concept check: 2. Correct. The field is the same and charge magnitudes match; opposite signs reverse force direction.

Written method: Magnitude = 0.003 times 200 = 0.60 N. Direction is west because the charge is negative. Two marks for magnitude and two for direction with reasoning.

Flux and Gauss law

Return to lesson: page 13.

Your turn: Yes. Field enters and leaves; the total closed-surface flux from that external charge cancels.

Concept check: 3. Correct. The crossing component decreases to zero as the normal becomes perpendicular to the field.

Written method: Parallel normal: flux = 50 times 0.20 = 10 N m squared/C. Perpendicular normal: cosine 90 degrees = 0, so flux = 0. Two marks per case.

Symmetry and conductors

Return to lesson: page 16.

Your turn: 20 N/C: the centre distance doubles and the field is divided by four.

Concept check: 1. Correct. Centre distance equals sphere radius plus the surface gap, so the two doubling operations differ.

Written method: Inverse-square scaling gives 80 times (0.20/0.40) squared = 20

N/C. Credit the outside-sphere condition and inverse-square reasoning.

Potential is not field

Return to lesson: page 19.

Your turn: ΔU = +6 J and work by the field = -6 J.

Concept check: 2. Correct. Charge times potential difference is negative for a negative charge and positive difference.

Written method: Energy change = 2 times 3 = +6 J. Work by field = -6 J. Two marks each; the signs are essential.

Capacitors and dielectrics

Return to lesson: page 22.

Your turn: 48 μC, from Q = CV.

Concept check: 3. Correct. The battery fixes voltage and supplies the extra charge associated with larger capacitance.

Written method: Q = CV = 4 times 12 = 48 microcoulombs. In SI this is 0.000048 C. Credit formula, substitution, prefix and result.

Current, resistance and electrical power

Return to lesson: page 26.

Your turn: Resistance doubles.

Concept check: 1. Correct. Both current and its corresponding voltage double, so their product quadruples.

Written method: V = IR = 6 V. P = VI = 12 W (or I^2R). Two marks per result.

DC networks, Kirchhoff rules and RC behaviour

Return to lesson: page 29.

Your turn: No. Once charged, it blocks steady DC in the ideal model.

Concept check: 2. Correct. At common current, voltage drop is proportional to resistance.

Written method: Total resistance = 12 Ω; current = 2 A; drops are 8 V and 16 V. One mark each; the drops must sum to 24 V.

Magnetic force: direction before calculation

Return to lesson: page 34.

Your turn: No. The angle is 0° or 180°, so its sine is zero.

Concept check: 3. Correct. At parallel alignment the perpendicular component and sine factor vanish.

Written method: F = BIL = 0.30 times 2 times 0.50 = 0.30 N. Four marks for formula, perpendicular condition, substitution and unit.

Sources of magnetic fields

Return to lesson: page 37.

Your turn: 10 μT because distance triples.

Concept check: 1. Correct. Doubling current and distance multiplies the numerator and denominator by the same factor.

Written method: B is inversely proportional to r, so B = 30 times 2/6 = 10 microtesla. Credit model, ratio, arithmetic and unit.

Induction and Lenz law

Return to lesson: page 40.

Your turn: No. Its magnetic flux is not changing.

Concept check: 2. Correct. Emf depends on the rate of flux change, so halving time doubles that rate.

Written method: Magnitude = 20 times 0.03/0.10 = 6 V. The direction needs the direction of the flux change; it cannot be inferred from these magnitudes alone.

Inductance, AC and electromagnetic waves

Return to lesson: page 43.

Your turn: 20 W: $V_{rms}^2/R = (20^2/2)/10$.

Concept check: 3. Correct. The heating depends on voltage squared, which is nonnegative for either polarity.

Written method: RMS = 20/square root of 2 = 14.14 V. Average power = RMS squared/10 = 20 W. Two marks per stage.

Electricity and Magnetism / Test A

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. A neutral object loses electrons. It becomes:

   A. Negative

   B. Positive

   C. Uncharged forever

   D. Magnetic only

2. A negative test charge in a field directed right feels force:

   A. Left

   B. Zero

   C. Right

   D. Up

3. A charge outside a closed empty box gives net flux through the box of:

   A. $\frac{q}{\varepsilon_0}$

   B. Always positive

   C. Always negative

   D. Zero

4. Inside the material of a conductor in electrostatic equilibrium, the electric field is:

   A. Equal to its surface value

   B. Infinite

   C. Zero

   D. Always outward

5. For equal and opposite point charges, take potential as zero at infinity. At their midpoint the potential is:

   A. Zero

   B. Undefined

C. Positive

D. Negative

6. Adding a dielectric to an isolated charged capacitor increases C. Its voltage:

A. Decreases

B. Increases

C. Must stay fixed

D. Becomes infinite

Part II / Show your reasoning

1. Two charges exert 12 N at separation r. What is the force at separation 2r, and at r if one charge is tripled?

2. A field of 200 N/C points east. Find the force on a charge of -0.003 C.

3. A uniform field of 50 N/C passes through a flat area of 0.20 square metres. The chosen surface normal points in the same direction as the field. Find flux. What if the normal is perpendicular to the field?

4. An isolated charged conducting sphere, with no external fields, has field magnitude 80 N/C at r = 0.20 m. Find the field at r = 0.40 m. Both distances are from the centre and outside the sphere.

5. A +2 C charge moves through a potential difference of +3 V. Find its potential-energy change and the work done by the field.

6. A 4 $\mu\mathrm{F}$ capacitor is connected to 12 V. Find the charge in microcoulombs.

Electricity and Magnetism / Test B

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. At fixed voltage, doubling resistance makes power:

   A. Unchanged

   B. Double

   C. Half

   D. Four times

2. After a long time in an ideal DC charging circuit, capacitor current is:

   A. Maximum

   B. Negative forever

   C. Zero

   D. Equal to the battery voltage divided by the resistor resistance

3. A charge moving parallel to a magnetic field experiences magnetic force:

   A. qvB

   B. Twice qvB

   C. Opposite its velocity

   D. Zero

4. Two parallel wires carrying currents in the same direction:

   A. Repel

   B. Attract

   C. Have no interaction

   D. Must carry zero field

5. Which situation produces an induced emf?

   A. No change of area, field or angle

   B. Changing magnetic flux

   C. A stationary unchanging field only

D. Constant flux

6. In vacuum, electric and magnetic fields of a plane EM wave are:

A. Parallel to each other

B. Longitudinal only

C. Zero

D. Perpendicular to each other and to travel

Part II / Show your reasoning

1. A 3 Ω resistor carries 2 A. Find voltage and power.

2. A 4 Ω and 8 Ω resistor are in series across 24 V. Find total current and each voltage drop.

3. A straight 0.50 m wire carries 2 A perpendicular to a 0.30 T field. Find force magnitude.

4. The field is 30 microtesla at 2 cm from a long straight wire. Find the field at 6 cm at the same current.

5. A 20-turn coil has flux per turn changing by 0.03 Wb in 0.10 s. Find average emf magnitude.

6. A sinusoidal voltage has peak value 20 V. Find its RMS value, then the average power in a 10 Ω resistor.

Electricity and Magnetism / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: B. Removing negative charge leaves a positive net charge.

Written solution: At 2r: 12/4 = 3 N. Tripling one charge at r: 36 N. Two marks for each ratio and result.

Return to lesson: page 6.

Question 2

MCQ: A. The negative charge reverses the field direction in $F = qE$.

Written solution: Magnitude = 0.003 times 200 = 0.60 N. Direction is west because the charge is negative. Two marks for magnitude and two for direction with reasoning.

Return to lesson: page 9.

Question 3

MCQ: D. No charge is enclosed, so total entering and leaving flux cancels.

Written solution: Parallel normal: flux = 50 times 0.20 = 10 N m squared/C. Perpendicular normal: cosine 90 degrees = 0, so flux = 0. Two marks per case.

Return to lesson: page 13.

Question 4

MCQ: C. Free charges redistribute until the internal field is cancelled.

Written solution: Inverse-square scaling gives 80 times (0.20/0.40) squared = 20 N/C. Credit the outside-sphere condition and inverse-square reasoning.

Return to lesson: page 16.

Question 5

MCQ: A. The signed scalar potentials cancel when distances and charge magnitudes match.

Written solution: Energy change = 2 times 3 = +6 J. Work by field = -6 J. Two marks each; the signs are essential.

Return to lesson: page 19.

Question 6

MCQ: A. Charge remains fixed and $V = \frac{Q}{C}$.

Written solution: Q = CV = 4 times 12 = 48 microcoulombs. In SI this is 0.000048 C. Credit formula, substitution, prefix and result.

Return to lesson: page 22.

Electricity and Magnetism / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: C. $P = \frac{V^2}{R}$, so doubling the denominator halves it.

Written solution: V = IR = 6 V. P = VI = 12 W (or I^2R). Two marks per result.

Return to lesson: page 26.

Question 2

MCQ: C. After a long time, the capacitor voltage equals the battery voltage. The resistor voltage and circuit current are zero. During charging, charge moves in the wires onto and off the plates; it does not cross the ideal dielectric.

Written solution: Total resistance = 12 Ω; current = 2 A; drops are 8 V and 16 V. One mark each; the drops must sum to 24 V.

Return to lesson: page 29.

Question 3

MCQ: D. The sine of the angle is zero.

Written solution: F = BIL = 0.30 times 2 times 0.50 = 0.30 N. Four marks for formula, perpendicular condition, substitution and unit.

Return to lesson: page 34.

Question 4

MCQ: B. Each current experiences the magnetic field of the other wire.

Written solution: B is inversely proportional to r, so B = 30 times 2/6 = 10 microtesla. Credit model, ratio, arithmetic and unit.

Return to lesson: page 37.

Question 5

MCQ: B. Faraday law depends on the rate of flux change.

Written solution: Magnitude = 20 times 0.03/0.10 = 6 V. The direction needs the direction of the flux change; it cannot be inferred from these magnitudes alone.

Return to lesson: page 40.

Question 6

MCQ: D. A plane electromagnetic wave in vacuum is transverse.

Written solution: RMS = 20/square root of 2 = 14.14 V. Average power = RMS squared/10 = 20 W. Two marks per stage.

Return to lesson: page 43.

Formula sheet

Charges, conductors and Coulomb force p.6

Rule

$$F = k|q_1 q_2|/r^2$$

$$k = 8.99 \times 10^9 \ \mathrm{N\,m^2/C^2}$$

Electric field and superposition p.9

Rule

$$E = F/q_{\text{test}}$$

$$E = k|Q|/r^2$$

$$\vec{F} = q\vec{E}$$

Flux and Gauss law p.13

Rule

$$\Phi = EA\cos\theta$$

$$\Phi_{\text{closed}} = Q_{\text{enclosed}}/\epsilon_0$$

Symmetry and conductors p.16

Rule

$$E_{\text{outside sphere}} = k|Q|/r^2$$

$$(r > R)$$

Potential is not field p.19

Rule

$V = kQ/r$ (zero at infinity)

$$\Delta U = q\Delta V$$

$$W_{\text{field}} = -\Delta U$$

Capacitors and dielectrics p.22

Rule

$$Q = CV$$

$$U = \tfrac{1}{2}CV^2$$

$$C_{\parallel} = C_1 + C_2$$

$$1/C_s = 1/C_1 + 1/C_2$$

Current, resistance and electrical power p.26

Rule

$$I = \Delta Q/\Delta t$$
$$V = IR$$
$$R = \rho L/A$$
$$P = VI$$

DC networks, Kirchhoff rules and RC behaviour p.29

Rule

$$R_s = R_1 + R_2$$
$$1/R_p = 1/R_1 + 1/R_2$$
$$\tau = RC$$

Magnetic force: direction before calculation p.34

Rule

$$F = |q|vB\sin\theta$$
$$F_{\text{wire}} = BIL\sin\theta$$

Sources of magnetic fields p.37

Rule

$$B_{\text{wire}} = \mu_0 I/(2\pi r)$$
$$B_{\text{solenoid}} = \mu_0 nI$$

Induction and Lenz law p.40

Rule

$$|\mathcal{E}| = N|\Delta\Phi_B|/\Delta t$$
$$\Phi_B = BA\cos\theta$$

Inductance, AC and electromagnetic waves p.43

Rule

$$U_L = \tfrac{1}{2}LI^2$$
$$V_{\text{rms}} = V_{\text{peak}}/\sqrt{2}\ \text{(sinusoidal)}$$
$$c = f\lambda\ \text{(vacuum)}$$

Symbol glossary

Symbols are listed with their lesson context. A symbol may represent different quantities in different rules.

Charges, conductors and Coulomb force

F electrostatic force magnitude / N

q_1, q_2 source charges / C

r separation of charges / m

Electric field and superposition

E electric field magnitude / N/C

q charge experiencing the force / C

F force magnitude / N

Flux and Gauss law

Φ electric flux / N m^2/C

E electric field magnitude / N/C

A surface area / m^2

Q_{enclosed} net charge inside a closed surface / C

Symmetry and conductors

Q net sphere charge / C

R sphere radius / m

r distance from sphere centre / m

E electric field magnitude / N/C

Potential is not field

V electric potential / V = J/C

ΔV final minus initial potential / V

ΔU change in potential energy / J

q moving charge / C

Capacitors and dielectrics

C capacitance / F

Q stored charge magnitude on either plate / C

V potential difference between plates / V

U stored electric energy / J

Current, resistance and electrical power

I conventional current / A

R resistance / Ω

V potential difference / V

P electrical power / W

ρ resistivity / Ω m

DC networks, Kirchhoff rules and RC behaviour

R_{eq} equivalent resistance / Ω

I current / A

V voltage / V

τ RC time constant / s

Magnetic force: direction before calculation

q particle charge / C

v particle speed / m/s

B magnetic field magnitude / T

F magnetic force magnitude / N

θ angle between velocity and field / degrees or radians

Sources of magnetic fields

B magnetic field magnitude / T

I wire current / A

r perpendicular distance from wire / m

n solenoid turns per unit length / m^{-1}

Induction and Lenz law

Φ_B magnetic flux through one turn / Wb

N number of coil turns / dimensionless

$\mathcal{E}$ induced emf / V

Δt time interval of flux change / s

Inductance, AC and electromagnetic waves

V_{peak} maximum sinusoidal voltage magnitude / V

V_{rms} root-mean-square voltage / V

L inductance / H

f wave frequency / Hz

λ wavelength / m

Source and coverage notes

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

Open original course folder

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Topic index

Find a lesson alphabetically. For individual symbols, use the subject symbol glossaries listed in the contents.

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